IDSS Linear Algebra In-Class Exercises

Iraklis A. Klampanos
University of Glasgow
✉ iraklis.klampanos@glasgow.ac.uk

16 October 2025

Note: *These practice questions have been designed to help you prepare for the IDSS exam, focusing on the Linear Algebra part of the module. Keep in mind that the actual exam questions will be different in content, style, and length, and they may even cover a different part of the taught material altogether. Please use these exercises to understand the material and practice the concepts and methods and do not memorise them.*

Questions

1. Assume the linear transformation described by a matrix M maps two points $P \rightarrow P'$ and $Q \rightarrow Q'$, such that $P' = MP$ and $Q' = MQ$.

   Explain how you would calculate the matrix M from the coordinates of these two points and their transformations using linear algebra. Explain your method and reasoning, also providing all relevant equations and matrices that you would use for this calculation.

2. Calculate M for $P = (5, 2)$, $P' = (-3, 4)$, $Q = (-1, -1)$ and $Q' = (2, 4)$.
   Hint: Use row reduction (transformation to a row echelon form, or even to a reduced row echelon form – RREF) to achieve this.

3. Solve the system above for $P = (2, 4), P' = (1, 1), Q = (4, 8), Q' = (2, 2)$. What can be said about the (linear transformation) matrix M in this case?

4. Assuming a square matrix $M = [m_{ij}] = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ show the relationship between m_{ij} of matrix M that indicate the M is singular.

   Hint: Start by assuming a linear dependence between the rows of M, which indicates that the implied system has no or not unique solutions.

5. The rank of a matrix is the maximum number of linearly independent rows or columns. It tells us the dimension of the row space or column space – that is, the number of directions in which the matrix contains meaningful information.

   Compute the rank of the matrix below, using row reduction:

$$A = \begin{bmatrix} 37 & -22 & 59 & 81 \\ 111/2 & -33 & 177/2 & 243/2 \\ -74 & 44 & -118 & -162 \\ 185/4 & -110/4 & 295/4 & 405/4 \end{bmatrix}$$

6. Given a zero-mean dataset represented by the matrix

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

compute the *whitening transformation matrix* W such that the transformed data $Z = WX$ has an identity covariance matrix.
Hint: Use the eigendecomposition of the covariance matrix.

7. Consider the covariance matrix

$$\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$$

Compute the whitening matrix W using eigen-decomposition.
(Whitening transformation – general case.)

8. Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

and interpret their geometric meaning.

9. Compute the singular value decomposition of

$$B = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$$

and verify that $B = U\Sigma V^\top$.

10. Prove that the eigenvalues of a diagonal matrix are the elements on its main diagonal. Then, explain how these relate to the singular values of the matrix.